Multi-Step Word Problems Without Parentheses

Multi-Step Word Problems • Standard: CCSS.MATH.4.OA.A.3

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Evaluate chained arithmetic expressions without parentheses by managing stack register flow.

In a multi-step expression like $(3 \times 4) + (2 \times 5)$, each product is a self-contained quantity that must be evaluated before the two groups can be added. On the 4-level stack, the first product (12) waits safely on Shelf Y while the second product (10) is formed on X, requiring zero parentheses.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Place Value Blocks & 4-Register Stack

Hands-On Setup: Use Place Value unit blocks to represent quantities alongside the 4-line stack display.

Student Prediction (Before Calculator): Problem: (8 - 3) × 2. Predict: 8 - 3 = 5, then 5 × 2 = 10.

3. CALCULATOR EXECUTION (SC-6 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [8] [ENTER] [3] [-] [2] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[8]	8	8 in X
[ENTER]	8	8 copied to Y
[3]	3	3 entered into X
[-]	5	8 - 3 = 5 in X
[2]	2	Automatic lift! 5 lifts to Y; 2 enters X
[×]	10	5 × 2 = 10 in X!

4. MATHEMATICAL CONTEXT & REAL-WORLD MOTIVATION (Inside-Out Multi-Step Problem Solving & Young Gauss's Pairing Insight)

In 1786, 9-year-old Carl Friedrich Gauss stunned his teacher by adding 1 through 100 in seconds—grouping the numbers from the outside in into 50 pairs of 101! Whenever we solve a multi-step story problem on paper, we naturally compute the inner group first and hold that intermediate result aside while we set up the next step.

Core Investigation (Student Warm-Up A):

A baker bakes 15 cookies, sells 6 of them, and packs the remaining cookies equally into 3 bags. How many cookies go in each bag?

Automatic Stack Lift & Register Retention

Order of Operations & Stack Lift • Standard: CCSS.MATH.4.OA.A.3

1. EXTENSION CONCEPT (WARM-UP B), BLOCKS & PREDICTION

Concept & Stack Approach: When an inner operation finishes, its result becomes the starting quantity for the next step. On the stack, that intermediate result automatically lifts to Waiting Shelf Y as soon as the next number is typed—allowing multi-step math like (12 ÷ 4) + 7 to flow continuously from the inside out.

Blocks & Prediction: Observe the 4-line stack screen as values automatically lift into register Y: (12 / 4) + 7. Prediction: 12 / 4 = 3. Entering 7 will lift 3 to Y. 3 + 7 = 10.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[1] [2] [ENTER] [4] [÷] [7] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1]	1	First digit of 12 enters X
[2]	12	12 in register X
[ENTER]	12	12 copied to Y
[4]	4	4 enters X
[÷]	3	12 / 4 = 3 in X
[7]	7	7 entered: 3 lifts to Y automatically!
[+]	10	3 + 7 = 10!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Pressing [ENTER] after an operator (+, −, ×, ÷) before typing the next number in a chained calculation, wasting a stack register.

Hands-On Block Remediation: Demonstrate Automatic Stack Lift: after [18] [ENTER] [6] [−] produces 12, typing [3] automatically lifts 12 to Register Y—no [ENTER] needed before [3] [×]!

Misconception 2: Reversing operand order in subtraction or division (computing X − Y instead of Y − X).

Hands-On Block Remediation: Remind students that Register Y is always the starting quantity on top (Row 1) and Register X is the operator's action (Row 0); if entered backward, press [x swap y] before [−] or [÷].

4. EXTENSION MATHEMATICAL INSIGHT

When an inner operation finishes, its result becomes the starting quantity for the next step. On the stack, that intermediate result automatically lifts to Waiting Shelf Y as soon as the next number is typed—allowing multi-step math like (12 ÷ 4) + 7 to flow continuously from the inside out.

The 4-Register Stack: Four Operations & Automatic Lift

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: 4-REGISTER STACK ANATOMY

- Four registers: T (top), Z (third), Y (second/operand), X (screen/bottom).
- ENTER duplicates X into Y and pushes older values up into Z and T.
- Two-number operators (+, -, ×, ÷) combine Y and X, dropping Z to Y and T to Z.

Key Sequence: `8` `ENTER` `3` `-` `2` `×`

Investigation 1: Solve (14 - 5) × 3 by finding the difference inside the parentheses first, then multiplying by 3.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 2: Solve 30 ÷ (2 + 3) by adding the divisor group (2 + 3) first, then dividing 30 by that sum.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 3: Evaluate (9 + 7) ÷ (8 - 4) by finding the numerator sum and denominator difference first, then dividing.

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Investigation 4: A farmer collects 20 eggs, sets aside 2 cracked eggs, and divides the rest equally among 6 cartons. How many per carton?

1. Block Model / Fraction Sketch:

2. Step-by-Step Paper Work:

Step 1 (Inner Group / Unit):

Step 2 (Next Operation):

Equation & Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-6

Unit 05: The 4-Register Stack: Four Operations & Automatic Lift — Answer Key

Complete solutions and SC-6 verification keystrokes for Grades 3–4

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): A baker bakes 15 cookies, sells 6 of them, and packs the remaining cookies equally into 3 bags. How many cookies go in each bag?

Core Solution: 3 cookies per bag ((15 - 6) ÷ 3 = 3).

Core Check Keys: [1] [5] [ENTER] [6] [-] [3] [÷]

Extension Challenge (Warm-Up B): Evaluate (24 ÷ 6) + 15 by solving the division group first and then adding 15 to that result.

Extension Solution: 19 (24 ÷ 6 = 4, then 4 + 15 = 19).

Ext. Check Keys: [2] [4] [ENTER] [6] [÷] [1] [5] [+]

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Solve (14 - 5) × 3 by finding the difference inside the parentheses first, then multiplying by 3.

Expected Answer: 27

SC-6 Verification Keys: [1] [4] [ENTER] [5] [-] [3] [×]

Investigation 2: Solve 30 ÷ (2 + 3) by adding the divisor group (2 + 3) first, then dividing 30 by that sum.

Expected Answer: 6

SC-6 Verification Keys: [2] [ENTER] [3] [+] [3] [0] [↑↓] [÷]

Investigation 3: Evaluate (9 + 7) ÷ (8 - 4) by finding the numerator sum and denominator difference first, then dividing.

Expected Answer: 4

SC-6 Verification Keys: [9] [ENTER] [7] [+] [8] [ENTER] [4] [-] [÷]

Investigation 4: A farmer collects 20 eggs, sets aside 2 cracked eggs, and divides the rest equally among 6 cartons. How many per carton?

Expected Answer: 3 eggs

SC-6 Verification Keys: [2] [0] [ENTER] [2] [-] [6] [÷]

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.